

Possible Hardware Causes of Overheating and How to Check Them

1. Faulty or blocked cooling fans

- **Cause:** The CPU or case fans may not be spinning properly, may be running slowly, or may have failed completely. This prevents hot air from being removed from the system.
 - **How I would check:**
 - Open the case and visually inspect the fans while the system is running.
 - Confirm that the CPU fan, GPU fan, and case fans are spinning normally.
 - Listen for unusual noises such as grinding or clicking.
 - Check fan speeds using hardware monitoring software.
 - **Resolution:** Replace faulty fans or improve airflow by ensuring all fans are functioning correctly.
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2. Dust buildup blocking airflow

- **Cause:** Dust accumulation on fans, vents, heatsinks, and filters can trap heat and prevent proper cooling, which is common in high-use environments like gaming cafes.
- **How I would check:**
 - Power off the system and open the case.
 - Inspect the CPU heatsink, GPU heatsink, fans, and air vents for dust buildup.
 - Check whether filters or ventilation areas are clogged.
- **Resolution:** Clean the system using compressed air, remove dust from components, and ensure proper airflow through the case.

3. Dried or improperly applied thermal paste

- **Cause:** Thermal paste between the CPU/GPU and heatsink transfers heat away from the processor. Old, dried, or poorly applied paste can cause temperatures to rise.
- **How I would check:**
 - Monitor CPU/GPU temperatures under load using monitoring software.
 - If temperatures are unusually high despite working fans and clean airflow, remove the heatsink and inspect the thermal paste.
 - Check whether the paste is dried, cracked, or unevenly spread.
- **Resolution:** Clean off the old thermal paste and apply a new layer before reinstalling the heatsink.

Additional check: Since the system is used for gaming, I would also verify that the GPU cooling system is working properly and that the power supply is not contributing to shutdowns under heavy load.